

The background is a composite image featuring medical scans, likely MRI or CT, of a human brain. The scans are rendered in a blue and white color scheme. Overlaid on the scans is a semi-transparent dark blue rectangle containing the title text. Additionally, there are various technical labels and data points scattered across the image, such as 'FoV 199', '296*512', 'Sag(1.5)', 'Tra>Cor(6.1)', 'AF', 'RFP', '5cm', 'R', 'FoV 199', '296*512', 'Sag(1.5)', 'Tra>Cor(6.1)', 'AF', 'RFP', '5cm', 'R', 'FoV 199', '296*512', 'Sag(1.5)', 'Tra>Cor(6.1)', 'AF', 'RFP', '5cm', 'R'.

Human In The Loop Simulation: Wizard of Oz

The image displays a collection of technical drawings, likely from a mechanical or structural engineering project. The drawings include:

- Top Left:** A drawing of a mechanical part with dimensions such as 17.5, 15, 12, 10.5, 8.1, 39, 7.5, 1.5, 32.5, 19, 7.5, 62, 200, 255, and 53. It also shows a cross-section view labeled 'A-A' with a 120-degree angle and a detail view labeled 'A(2:1)'.
- Top Right:** A drawing of a mechanical part with dimensions such as 17.5, 15, 12, 10.5, 8.1, 39, 7.5, 1.5, 32.5, 19, 7.5, 62, 200, 255, and 53. It also shows a cross-section view labeled 'A-A' with a 120-degree angle and a detail view labeled 'A(2:1)'.
- Bottom Left:** A drawing of a mechanical part with dimensions such as 17.5, 15, 12, 10.5, 8.1, 39, 7.5, 1.5, 32.5, 19, 7.5, 62, 200, 255, and 53. It also shows a cross-section view labeled 'A-A' with a 120-degree angle and a detail view labeled 'A(2:1)'.
- Bottom Right:** A drawing of a mechanical part with dimensions such as 17.5, 15, 12, 10.5, 8.1, 39, 7.5, 1.5, 32.5, 19, 7.5, 62, 200, 255, and 53. It also shows a cross-section view labeled 'A-A' with a 120-degree angle and a detail view labeled 'A(2:1)'.

- [illegible]

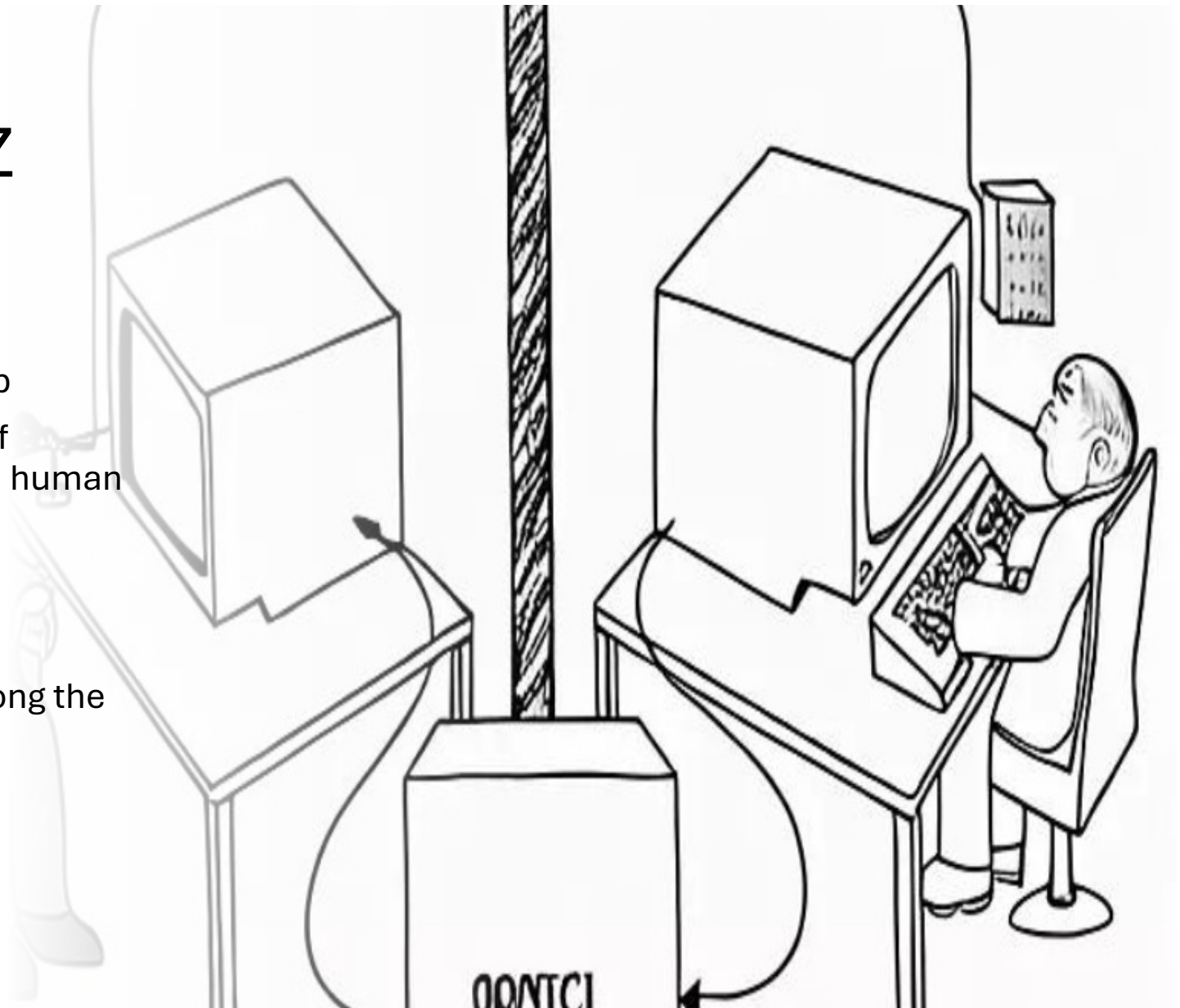
USING simulator

- Flight, Driving, Management simulators and so on can be used to immerse users
- Allows iterative design, enhancement of existing systems or to test the deployment of new procedures
- Sandbox version of some systems (ex. Moodle)



WIZARD of OZ experiment

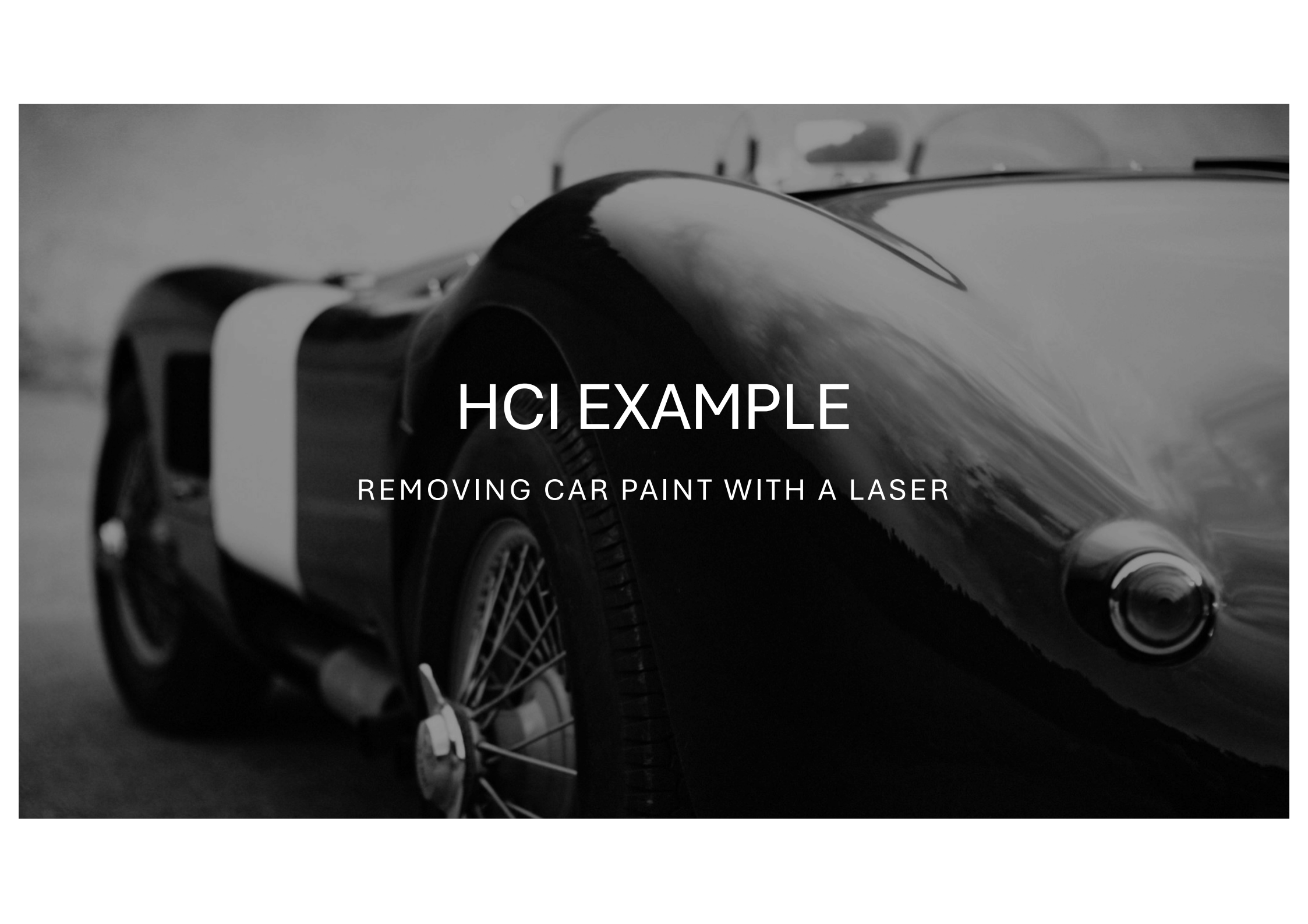
- Experimenter-in-the-Loop
- Simulate the behaviour of intelligent systems with a human agent behind the scene
- Iterative prototyping process, replace human interventions with system functions along the way
- [Kelley. 2018]



Principle of the wizard of Oz

- Cost-effective alternative when technology development is prohibitively expensive
- Useful for focusing and narrowing down problem spaces
- Useful for:
 - Assessing interface utility
 - Pushing system boundaries
 - Identifying potential user behavior
 - Resource-efficient exploration before full technical solution development





HCI EXAMPLE

REMOVING CAR PAINT WITH A LASER

HCI Example



Little Intensity

HCI Example



An aerial photograph of a large offshore oil platform in the middle of a dark blue sea under a cloudy sky. The platform is a complex of yellow and grey metal structures, including a large crane on the left and a tall derrick on the right. A long yellow walkway leads from the bottom center towards the platform. The text 'HSI EXAMPLE' is overlaid in large white letters, and 'DESIGN OF REMOTE OPERATION FOR OFFSHORE OIL PLATFORMS' is overlaid in smaller white letters below it.

HSI EXAMPLE

DESIGN OF REMOTE OPERATION FOR OFFSHORE OIL
PLATFORMS

Réunion du matin



Dans la salle de contrôle/salle d'opérations



- 1 opérateur robot
- 1 paneliste
- 1 superintendant



- Liste des permis de la journée
- Permis de travail de l'intervention de raclage (avec PID : Plan Isolation Des vannes)

Terrain



1 personne sur le terrain Magicien d'Oz



Webcam



Robot Tracker



Equipements sur TADI

Téléopération

Liaison par
radio

Salle de contrôle / salle d'opérations



Xbox Game Barre



1 PC avec vue direct du terrain



1 PC Jumeau Numérique avec
modélisation 3D de TADI sous Unity



Xbox Game Barre



1 PC Station de commande du robot avec
logiciel TAURUS



Xbox Game Barre



1 PC avec écran secondaire Station de contrôle du
procédé avec document Power Point

Human in the loop simulations Wizard of Oz



1 expérimentateur magicien d'Oz

1 opérateur robot



GoPro



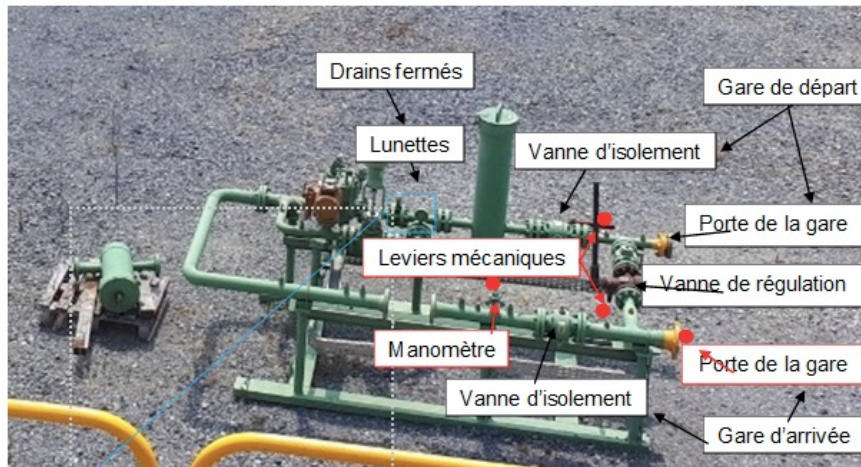
1 paneliste



GoPro

Human in the loop simulations

Physical and digital twin



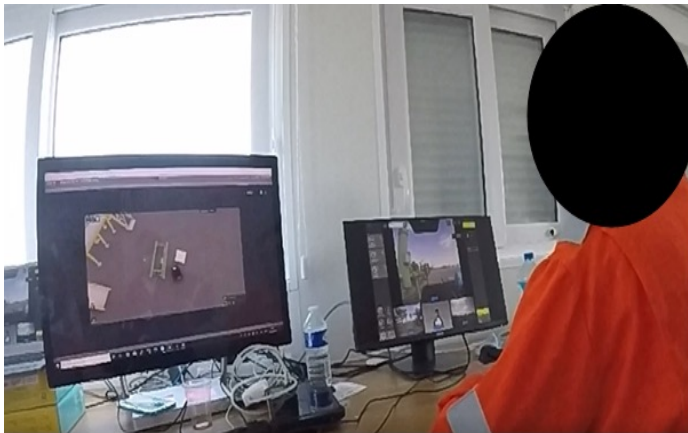
Fake installation to simulate a pig launcher



Digital twin of the installation and robot

Human in the loop simulations

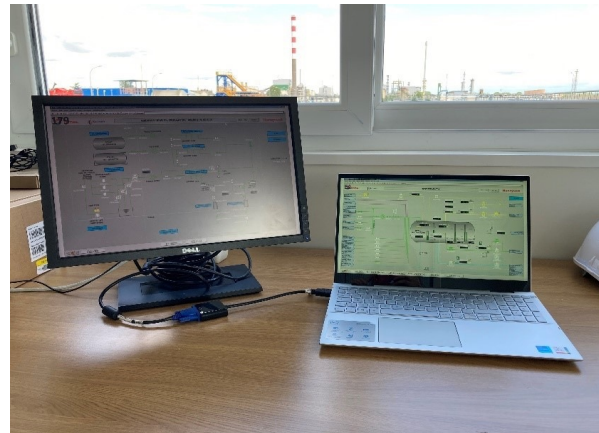
User interfaces and resources



Operation room :

Digital twin (left)

robot remote controler (right)

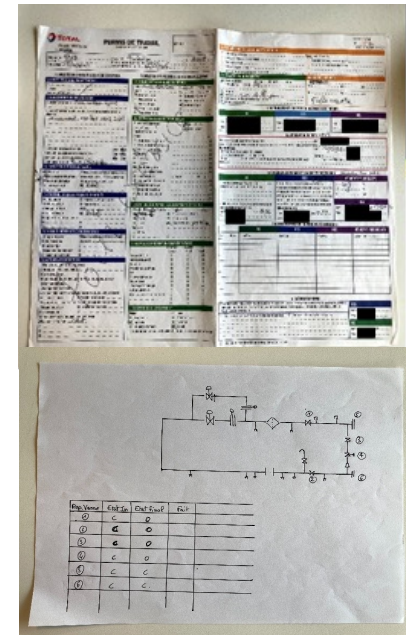


Control room :

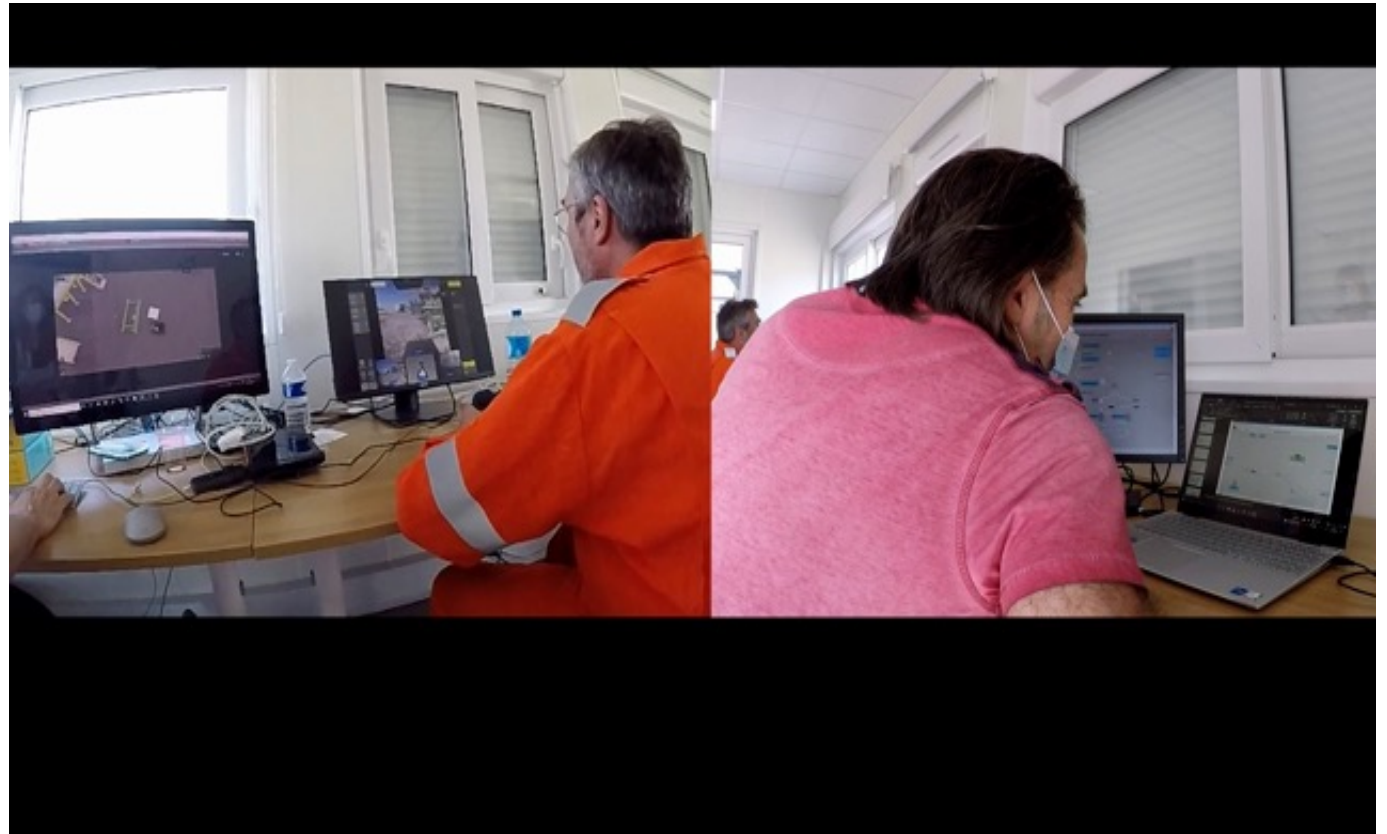
Process controller (powerpointware left)

Work permits (right top)

Process Description (right bottom)



HCI Example



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